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LIMITATIONS SUPPORT LEARNING: A DESIGN BASED STUDY OF ETHICALLY SCAFFOLDED GENERATIVE AI TOOLS FOR LONG FORM ARGUMENTATION AND METACOGNITION

by

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Master of Arts in Curriculum and Instruction, University of Central Florida, 2009

A dissertation submitted in partial fulfillment of the requirements
for the degree of Doctor of Education
in the Department of Learning Sciences & Educational Research
in the College of Community Innovation and Education
at the University of Central Florida
Orlando, Florida

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Major Professor: David N. Boote

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ABSTRACT

This design-based research study implemented and refined an instructional model that used generative AI (GAI) tools within a nine-week AP Capstone Seminar argumentation unit. The model combined teacher feedback and modeling, structured reflection tasks (including AI-assisted reflection), AI assisted editing, AI-assisted literature searches, and counterargument support using restricted chatbots. During the nine week period, responsibility for feedback moved from teacher and AI supported scaffolding to student driven writing practice and revision. Data collection focus on four advanced high school students at a Florida public high school, including multiple drafts, GAI input and output logs, reflections, and pre- and post-unit semi-structured interviews.

Data analysis showed that with these scaffolds in place, students improved their counterarguments, evidence integration, and essay structure. However, counterargument and rebuttal were not integrated with the main line of reasoning. As students became more self regulated and used more nuanced prompts with the GAI tools, metacognitive growth emerged. The tools supported this growth by prompting students to plan, monitor, and evaluate their writing choices, especially during revision and reflection tasks. AI-assisted reflections and editing tools gave students immediate feedback that encouraged intentional reconsideration of structure, evidence use, and argumentative clarity. In this model, limitations such as restricted tool access, scaffolded reflection, and gradually withdrawn support became productive constraints that helped students build independence and insight. Ethical awareness changed from naive acceptance to a more discerning evaluation of each tool's benefits and limits. Over time, reliance on AI tools decreased, indicating both improved writing ability and strategic use of technology. Teaching students to question and direct AI output is as important as traditional writing instruction and improves their metacognition and ethical use of AI in high school writing classrooms.

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Dedicated to those who “believed I could”

To my mother and father, who always encouraged me to work hard and follow my dreams; whose strength, love, and unwavering support of not just my education but to my pursuit of happiness, paved the way to belief in myself.

To my daughter and son, my whole heart…who inspire me every day to keep going. Because of them I know that I can accomplish anything.

“She believed she could…so she did.”

-R.S. Grey

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"There will be a few times in your life when all your instincts will tell you to do something; something that defies logic, upsets your plans, and may seem crazy to others. When that happens, you do it." -Judith McNaught

McNaught must have been talking about writing a dissertation. She *had to* be.

Living here in the theme park capital of the world, it might surprise you to know I've never been a fan of thrill rides, the looping, twisting, and turning, the motion sickness, the vertigo. The only time I could stomach it (pun intended) was when I had a group of people that wanted to see me "conquer the beast" around me as moral support. This dissertation is the culmination of a journey that if you had asked five years ago, I would have never said I would start…but boy have the past three years been a ride, looping and twisting and all over the place…and I never would have made it through without the amazing support I've received in this program, so some thanks are definitely in order.

To the CCIE program and its brilliant faculty, thank you for turning me into a researcher, not just someone with a tendency to go down intellectual rabbit holes and call it productivity. Many of you were integral in preparing me for this moment.

To my amazing committee, I wish to impart the sincerest of "thank you's" that have ever been thanked. Thanks to all of you for meeting with me to discuss and plan the approach to this monster of a project. To Dr. Ceballos for her incredible understanding of what it's like to be an AP English Language teacher, who was intimately familiar with the writing struggles our high schoolers face nowadays, her insights and suggestions were right on target and so helpful. To Dr. Campbell, for accepting just one more committee request, for coming through in the eleventh hour and being open to the possibilities of using generative AI systems for good, and not for evil. You are truly a gem. To Dr. Taub for being so welcoming and so patient in teaching me how to "think about my thinking" and bolstering my resolve hundreds of times when I was scared that I would not find the data that I needed...you were right, it was in there. And to Dr. Gunter, who could not join us at the party, but lent tremendously valuable insight, you are

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amazing…and I hope you are enjoying the well deserved time off! All of you set a standard to aspire to, and I am incredibly lucky to have met and worked with you all.

To my advisor and chair Dr. Boote, given my usual verbosity, there are shockingly few words that I can pull from my lexicon that seem adequate to express my gratitude, but I'll give it the old college try. You may have heard me say once or twice that I am not a "conventionally lucky" person…I've never won the lottery, or drawn a royal flush, and my bingo game is downright abysmal. My luck is instead "unconventional"; the first parking space at the supermarket, or the day Sabrina met Anna and Elsa by happenstance at Epcot on her sixth birthday, saving us a wait in a seven hour line and a meltdown of epic proportions. And that's the luck that came into play the day I met you. Regardless of my capacity to be a PITA, thank you for being patient and supportive, reminding me that I could see this through, even on days where I wasn't sure that I could. Thank you for picking me up and dusting me off and telling me I was "exactly where I needed to be" when my brain was going haywire through data analysis, or a million other instances. Thank you for your sense of humor, and laid-back approach, that allowed me to never be scared to tell you what I thought honestly. Thank you for being open to my opinions. Thank you for giving the best advice, answering thousands of questions (many more than once), and ensuring that I forever and always refer to rough drafts as "SFD's". Thank you for putting up with my tears empathetically on days when I hit Defcon ONE, and on other days when I was less than a ray of sunshine, thank you for being able to make me laugh with one of your stories. Thank you for reminding me to not sweat the small stuff. Thank you for being welcoming when my children or my husband had something to say that *just wouldn't wait*, crashing Zoom meetings and allowing them to sit in on classes. Thank you for checking in on all of us when we hit unexpected bad news. Thank you for being a dog person…*it matters*. Mostly, thank you for helping me realize exactly what type of teacher I want to be from here on out. I know that your life is a constant stream of new students, that's just the nature of our business, you know? However, because of that constant stream of faces it gets increasingly difficult to remember some of them after a while, and I expect I will be no different. Just know that while I may soon fade from your memory, you've left an indelible mark on mine. There is so much more I could say, but this paper is already long

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enough so just…from the bottom of my heart…thanks and appreciation and ALL of the confetti… in perpetuity.

To my family, I could not have done this without you. Thanks to my mom and dad, my own personal heroes and cheerleaders for supporting me, for listening to me talk for hours on end about concepts they admittedly did not understand, for showing me my worth and capabilities, and reminding me that I can do anything…not just because *"ego sum corporis,"* (although I'm sure it certainly helps) but because I am strong, smart, and stubborn enough to persevere. I love you both so much. Thank you to my husband Brian for enduring this journey with aplomb and ensuring that I always had time to attend class, analyze data, and type thousands of words. To my incredible, smart, loving, and gorgeous (albeit at times obnoxious) children, you are the reason I get up in the morning. You make me want to be better: a better mom, a better role model…just all around *better.* I'm sorry for all the times I wasn't paying attention because I was writing or analyzing data and you wanted to talk about something, but I wanted you to see that there is *nothing* out of your reach. If mama can go back to school and become a doctor, then there's no limit to what you can do. I wanted you to see the value of hard work, because not much that is good is ever easy. I love you "3000" and I can't wait to see what you will achieve.

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Best. Ride. *Ever.*

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LIST OF ABBREVIATIONS

AI: Artificial intelligence

AP: Advanced placement

AWE: Automated writing evaluation

DBR: Design-based research

GAI: Generative Artificial Intelligence

IMP: Individual multimedia presentation

IRR: Individual research report

IWA: Individual written argument

LOI: Line of Inquiry

NAEP: National Assessment of Educational Progress

NLP: Natural language processing

TMP: Team multimedia presentation

SC 1/2/3/4: Status check

SC1/2/3/4REV: Revised status check

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CHAPTER ONE

The rise of generative artificial intelligence (GAI) systems and their implications for education has pointed out a pressing need to investigate their potential use to promote student preparedness in a post-secondary setting. While these systems have some potential to disrupt authentic learning through academic dishonesty, their increasing prevalence cannot be ignored. By scaffolding the use of GAI tools to improve argumentative practice, students could gain valuable metacognitive skills, with real world applications, such as planning and monitoring their own reasoning, considering alternate perspectives, and self-regulation during the writing process. The development of these capabilities, obtained with practice and structure in the ethical and responsible use of generative artificial intelligence tools could ultimately help students develop and strengthen their argumentative writing proficiency.

Additionally, metacognitive growth and argumentative writing proficiency have long been intertwined. Arguments require students to engage in many higher order thinking skills such as evaluation, analysis, synthesis and effective communication, which are focus points of an advanced placement curriculum like the one found in AP Seminar coursework. The incorporation of various GAI tools to scaffold the learning of new skills for these types of students may in turn enhance their metacognitive abilities twofold; through engagement, writing and reflection on the Seminar curriculum, as well as critical evaluation of the GAI tools for identification of potential bias or inaccuracy, and use both to refine their argumentation accordingly.

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Problem statement

The problem that is the focus of this study is students' underdeveloped argumentative reasoning and writing skills in their long-form academic arguments through their lack of metacognitive growth, as well as the challenges educators face in teaching these complex skills well. Additionally, I understand that researcher positionality and inherent biases may influence the approach to this problem, so a detailed positionality statement addressing these factors can be found near the end of Chapter 3.

High school writing instruction systemically overlooks building foundations for postsecondary readiness: the ability to produce coherent, evidence-based, long form arguments (Lavell et al., 2002; Miller et al., 2018;). Rigorous argumentation instruction for students, including those considered advanced, remains a fundamentally overlooked area (Istiana et al.,2020; Klopp & Stark, 2020; Osborne et al., 2013). Without sufficient practice cultivating core argumentative writing skills, students' competency gaps increase by graduation (Graham, 2019). Crafting good arguments requires strategic integration of multiple elements: formulating a defensible thesis statement grounded in credible evidence, analyzing concepts through clear and logical reasoning, acknowledging multiple perspectives, addressing counterarguments, tailoring communication for specific audiences and contexts, and revising one's position based on newly acquired evidence (Butler & Britt, 2011; Graham, 2019).

Proficient argumentative writing can strengthen important cognitive skills of evaluation, perspective, and drawing conclusions (Billing, 2007; Soysal, 2021) through the construction of evidence-based arguments. Although not a panacea, through practice, the higher order thinking necessary for effective argumentation can transfer to improved analysis (Liu & Stapleton, 2014). However, thinking critically requires a thorough understanding of topics, and not just skills acquisition (Willingham, 2007). Gaps in argumentation instruction and practice can limit development of problem-solving abilities that are vital for future success, regardless of discipline.

However, typical secondary ELA curricula currently concentrate on personal narrative, literary critique, content based (as opposed to skills based) writing, and formulaic essay formats rather than cultivating complex rhetorical skills for persuasive writing (Butler & Britt, 2011). Additionally, teachers who rely exclusively on conventional instructional methods in which the teacher is more active and the students are more passive (Ka-Kan-Dee & Kaur, 2015), often labor to engage students and stimulate active participation in the writing process, leading to less participation from less engaged students. Undifferentiated writing assignments that fail to target skills specifically and generic, slow feedback

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exacerbate the problem and can lead to deficits in students' development in argumentative writing (Graham,2019; Kiuhara et al., 2009).

For this study, I chose to focus on long form argumentative writing, which consists of an extended argumentative assignment of 1,200 to 1,320 words, or approximately five pages, because it aligns with postsecondary preparedness in terms of an academic research paper. I am frustrated with how infrequently this type of writing is taught at the secondary level, with no mandatory research and argumentation requirement for high school graduation in the state of Florida (Florida Department of Education, n.d.).

These gaps in argumentative reasoning weaken high school graduates' readiness for postsecondary coursework (Lv & Chen, 2010; Newell et al., 2011; Scager et al., 2014; Venezia & Jaeger, 2013). Students who graduate high school but toil to persuasively share opinions on complex topics struggle in college classes (Coffelt et al., 2019) as well. Likewise, lacking exposure to elements of persuasive writing during secondary schooling can hinder career advancement. Employers prioritize clear communication skills to convey ideas or proposals, and without solid foundations in structuring cogent arguments from their K-12 education, graduates will find limited opportunities to showcase talents needed for professional growth. Strong argumentation involves integration of multiple elements, some of which involve metacognitive skills such as acknowledging multiple perspectives, addressing counterarguments, and revising one's position based on newly acquired evidence (Butler & Britt, 2011). It is imperative that educators seek new avenues to address these deficits now, particularly in developing students' metacognitive abilities for argumentative reasoning in their long form argumentative writing.

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Significance statement

Writing ability promotes achievement in schooling, careers, and society (Bangert-Drowns et al., 2004) and deficits in composition skills prevent students from reaching their full potential (Coker & Lewis, 2008). In high school and postsecondary education, grades in many courses are heavily

influenced by students' ability to express their thoughts clearly on written assignments (Graham & Harris, 2006). Composition skills also support learning of course content; writing serves as the means through which students clarify their understanding of concepts and retain knowledge (Bangert-Drowns et al., 2004; Graham & Perin, 2007). Either after or instead of postsecondary education, employability and promotion in the modern workforce depends on the ability to write clear reports, technical resources, emails, and business correspondence (Gillespie et al., 2013). Foundational writing skills allow people to actively help form outcomes, enabling an individual voice to directly influence a public domain. People who are otherwise very capable but lack proficient argumentation skills enter higher education and workplace environments at a disadvantage, and poor persuasive writing skills disrupt the potential for success in academics, corporate, and public sector careers that require effective communication.

Many high school students in Florida cannot articulate and understand complex arguments in postsecondary education and professional settings after graduation, due to a deemphasis on this type of writing instruction. National assessment data reveals only 24% of 12th graders were able to write proficient arguments (NAEP, 2012). This was no anomaly. Over a 22-year period, writing performance on the NAEP only fluctuated a few percentage points nationally, and Florida is no exception. More specifically, secondary students lacked ability to master general skills of academic writing like making evidence-based claims, addressing counterclaims, using credible sources strategically, and revising ideas based on new learning (Graham, 2019). Without tailored instruction and opportunities to practice core aspects of argumentation, deficits occur in thinking and problem-solving skills (Graham, 2018).

As such, this study addresses a perennial challenge faced by high school composition educators: secondary students' preparation for long-form, written academic arguments (Lin & Mintzes, 2010). In addition to the skills needed for all academic writing, argumentative writing also requires logical reasoning, providing evidence to support claims, anticipating counterarguments, conveying claims persuasively, sequencing arguments, and addressing multiple and divergent audience perspectives. These

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skills, and the concepts that support them, are major barriers for high school students (Kiuhara et al., 2009). Extensive research has examined the reasons why students wrestle with argumentative writing such as lack of instructional scaffolds and models (Jonsson, 2013), insufficient opportunity to practice (Graham & Perin, 2007), and inadequate or infrequent feedback (Applebee & Langer, 2011; Sharif & Zainudin, 2017).

The rise of generative AI availability and capability exposes another barrier. Although technology integration has proven time and time again to have effects on student learning that can be beneficial when used appropriately, and detrimental when not, these systems require a grasp of ethics and responsibility often overlooked in education due to their newness. The findings from this study could provide insights into the ways that GAI tools can be used as intended: instructional scaffolds and supports to develop cognitive and metacognitive skills in argumentative writing.

Proficiency in argumentative writing requires several cognitive and metacognitive skills, like anticipating counterarguments, evaluating evidence for relevance and credibility, crafting rebuttals, and monitoring one's own reasoning as writing and subsequent revision take place. Without adequate and appropriate scaffolding, students do not employ the *appropriate* metacognitive knowledge, skills, and strategies needed to construct and support arguments and, as a result, may disengage altogether (Kim et al., 2021). The significance for this study exists in it's potential to harness the power of instructional technology as a tool to enhance metacognitive ability *and* argumentative writing skill, as well as the development of pedagogical strategies for other teachers, enhanced writing curriculums that promote ethical and responsible use of large language models, as well as feedback options for students or teachers who are not yet confident in writing, or the teaching of it.

The implications extend beyond the immediate context of argumentative writing in advanced English Language Arts classes. The skills and strategies developed through this GAI-assisted approach could be helpful for students across different academic disciplines. History classes might apply these argumentative writing skills to construct better historical analyses. Science students could use these

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